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Article

Finite-Element Modeling of Additively Manufactured Regolith-Filled Polyether–Ether–Ketone (PEEK)

Mohammad Azami, Pierre-Lucas Aubin-Fournier and Krzysztof Skonieczny

Department of Electrical and Computer Engineering, Concordia University, Montréal, Québec, Canada
* Correspondence: mohammad.azami@mail.concordia.ca

Highlights

- Defect-free FE modeling predicts the upper-bound elastic modulus of PEEK/regolith composites.
- Microscopy-informed defect models quantify stiffness losses from pores and interfacial flaws.
- The effects of regolith loading and manufacturing defects are decoupled using FE simulations.
- Modulus drop at 50 wt% loading is consistent with defect connectivity in printed composites.
- The framework provides transferable insight for particle-reinforced high-performance thermoplastics.

Abstract

Additive manufacturing of PEEK/regolith composites offers a promising route for Lunar in-space manufacturing by reducing dependence on off-site material supply. However, the processability of these materials and the elastic response of printed components are strongly affected by regolith loading and manufacturing-induced defects. This study develops a hierarchical finite-element framework to quantify the stiffness of additively manufactured PEEK containing 10–50 wt% LMS-1D Lunar regolith simulant and to separate intrinsic composition effects from defect-driven losses. The approach combines composition-based estimation of regolith properties with microstructure-informed simulations of PEEK/regolith composites. Under defect-free assumptions, the predicted modulus increases monotonically from 1.27 GPa at 10 wt% to 1.97 GPa at 50 wt%, showing good agreement with experimental trends up to 40 wt%, where deviations remain within 3.5–10.1%. At 50 wt%, however, the experimental modulus decreases to 1.27 GPa, whereas the defect-free model predicts 1.97 GPa. Microscopy-informed single-layer analyses indicate that tall crack-like interfacial voids, polymer-starved welds, and interconnected weak seams significantly reduce load transfer and shift the mechanical response toward an interface-controlled regime. These results demonstrate that regolith additions can enhance stiffness only until defect connectivity becomes dominant. The findings provide insight into the process–structure–property relationships governing ceramic particle–reinforced high-performance thermoplastics in additive manufacturing.

Keywords: in-space manufacturing; Lunar-based manufacturing; additive manufacturing; polyether–ether–ketone (PEEK); finite element modeling

1. Introduction

Sustained human activity on the Moon will require manufacturing strategies that reduce dependence on Earth-supplied mass and convert local materials into usable feedstocks for fabrication, repair, and construction. NASA's Lunar in-situ resource utilization (ISRU) roadmap explicitly identifies regolith-derived feedstocks and in-space manufacturing as enabling capabilities for long-duration surface operations and future expansion beyond Earth [1–3].

Additive manufacturing is particularly attractive in this context because it enables mission-specific parts with limited tooling, reduced material waste, and substantial geometric freedom. Recent reviews have also identified regolith-based additive manufacturing as a major research direction for Lunar

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infrastructure and surface systems. Extrusion-based approaches are an attractive option because they can process composite feedstocks while avoiding some of the thermal demands and process complexity associated with fully molten ceramic routes [3–5].

Among candidate matrices, high-performance thermoplastics such as the polyaryl–ether–ketone (PAEK) family that includes Ppolyether-ether-ketone (PEEK) and Polyether-ketone-ketone (PEKK) are promising for Lunar applications. Additively manufactured PEEK has been widely recognized for lightweight, chemically stable, aerospace-relevant components. However, as a semi-crystalline thermoplastic, PEEK is challenging to print, and the mechanical response of printed PEEK is strongly influenced by process-induced features such as porosity, thermal history, and interlayer bonding. As a result, structural qualification is more complex than for conventionally processed materials [6,7].

These challenges become even more critical when PEEK is combined with Lunar regolith simulants to reduce the fraction of Earth-supplied polymer in the final component. Recent studies on regolith-filled PEEK and PEKK demonstrate that relatively high regolith loadings can be achieved in additively manufactured composites [6,8]. At the same time, experimental investigations across regolith–polymer systems indicate that increasing regolith content affects printability, stiffness, and dimensional stability, while also introducing compromises in strength, ductility, and processing windows [6,9].

For example, Azami et al. [6] successfully incorporated 10–50 wt% LMS-1D Lunar mare simulant into PEEK and subsequently fabricated the composites using additive manufacturing. The incorporation of regolith resulted in a gradual increase in elastic modulus, with improvements ranging from 6–41% for composites containing 10–40 wt% filler, respectively. However, this trend changed at 50 wt%, where the modulus was only 16% higher relative to pure PEEK and decreased by approximately 25 percentage points compared to the 40 wt% composite. These results represent a substantial improvement over their earlier work [10,11] following fine-tuning of the processing parameters and printing conditions. In particular, the 30 wt% composite demonstrated an improvement of more than 21.7% compared to the previous study and about 19.3% relative to pure PEEK. The underlying mechanisms were further interpreted through scanning electron microscopy (SEM) analysis, which revealed differences in microstructural features between the two phases of the study.

Similarly, Malekpour et al. [8] incorporated 10–60 wt% Lunar regolith simulant into a PEKK matrix and fabricated the composites using fused filament fabrication (FFF). Their results showed that annealing induces a substantial modulus increase of approximately 53% in neat PEKK. However, the addition of regolith significantly alters this thermomechanical response. For low-to-intermediate filler loadings (10–30 wt% LRS), annealing paradoxically reduces stiffness, resulting in a modulus decrease of approximately 22–24% compared to the as-printed condition. As the filler concentration increases to 50 wt%, annealing again produces a positive stiffening effect (approximately 10% increase). This trend eventually plateaus at 60 wt% LRS, where both printed and annealed specimens exhibit a comparable maximum modulus of nearly 1.7 GPa. Additional studies on additively manufactured thermoplastic–Lunar regolith composites have been reported in the literature. However, because the polymer matrices employed in those works do not belong to the PAEK family, they fall outside the scope of the present review (see [2,12]).

Although these studies have improved understanding of processing and experimental performance, the mechanics-based modeling of additively manufactured regolith–PEEK composites remains comparatively limited, particularly with respect to manufacturing-induced defects and their influence on effective mechanical response. This motivates the use of finite-element modeling to separate intrinsic composition-driven behavior from performance losses associated with defects introduced during fabrication [13].

A rigorous understanding of the intrinsic mechanical capability of PEEK–regolith composites is needed to distinguish composition-driven stiffening from performance losses introduced during additive manufacturing. Without a defect-free reference, the measured elastic response reflects not only regolith loading, but also porosity, weak interlayer welds, and particle–matrix interfacial defects, making it difficult to define clear processing targets and reliable design allowables.

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To address this gap, the present study uses finite-element modeling to establish the upper-bound defect-free elastic response of PEEK–regolith composites over 10–50 wt% regolith in 10% increments. The framework then incorporates microscopy-informed porosity and representative interfacial flaws to quantify the stiffness reduction associated with specific manufacturing defects. In this way, the model separates intrinsic material potential from defect-induced knockdown and clarifies how microstructural imperfections govern the departure of printed parts from their ideal response. This constitutes the central novelty of the work: a defect-aware, microstructure-informed modeling strategy that links composition, defect architecture, and elastic performance in additively manufactured PEEK–regolith composites, while providing a transferable methodology for other particle-reinforced high-performance thermoplastic systems.

2. Geometric Modeling of Composite

The first step in this analysis is to create a material model with varying regolith contents that includes both the matrix and particles while excluding structural defects. This establishes the idealized mechanical potential under flawless conditions. A representative volume element (RVE) of the composite is generated in CAD, followed by assignment of boundary conditions and material properties. The RVE is meshed and subjected to uniaxial tension at a constant nominal strain rate, mirroring the experimental setup from [6] to characterize elastic behavior. Plastic deformation is outside the scope of this study; we focus on the elastic regime to determine Young's modulus.

The analysis of idealized PEEK–regolith composites involves simplifying assumptions to keep the problem computationally tractable. First, additive manufacturing process effects are neglected; in particular, anisotropy and inhomogeneity from layer-by-layer fabrication are ignored, and the material is treated as isotropic and homogeneous. Second, the semi-crystalline nature of PEEK is simplified by neglecting spatial variations in crystallinity between amorphous and fully crystallized regions. Third, the morphology of regolith particles is idealized as random polygons, despite their inherently irregular shapes. Fourth, each particle is treated as homogeneous with a single effective (composition-averaged) elastic modulus, while in reality, regolith grains are multiphase aggregates with spatially varying properties. Finally, nanoscale effects are excluded because continuum finite element tools such as ABAQUS and ANSYS cannot capture quantum-scale interactions or surface-dominated phenomena. These assumptions enable practical modeling, but limit fidelity to real composite behavior.

Two-dimensional (2D) modeling is adopted as the most practical approach. The high particle count, complex morphologies, and broad size distribution render 3D models computationally prohibitive; trial runs show that a $4 \times 4 \times 4$ mm RVE produces CAD files on the order of tens of gigabytes, exceeding available resources. This choice is justified on two grounds. First, scanning electron microscopy indicates a reasonably homogeneous particle distribution, so an in-plane representation captures the essential microstructural statistics for elastic response. Second, the elastic tensile behavior of interest is well represented by planar stress–strain relationships; out-of-plane interactions and plasticity are not analyzed here, so a 3D model would add cost without commensurate insight. The 2D approach therefore balances computational efficiency with the study objectives.

Generating CAD Models of the Particles

The particulate geometry is generated with a MATLAB script. We first construct a 3D stochastic RVE populated with non-overlapping particles, then extract a 2D mid–plane section to obtain the planar CAD used for analysis. Under standard stereological assumptions, the areal fraction measured in the 2D section provides an unbiased estimate of the 3D particle volume fraction. User inputs include the maximum particle size, the matrix (RVE) dimensions, and the target particle volume fraction.

For robust control of volume fraction we seed the 3D RVE with spherical particles and later replace each sphere by a cube having the same circumscribed diameter. If the diameter of the (temporary) sphere is d, the edge length of the cube is $a = d/\sqrt{3}$ (since the circumscribed sphere of a cube has diameter $a\sqrt{3}$).

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The corresponding single-particle volumes are

$$Volume_{\text{sphere}} = \frac{\pi}{6} d^3, \qquad Volume_{\text{cube}} = a^3 = \frac{d^3}{3\sqrt{3}}. \tag{1}$$

so that

$$\frac{Volume_{\text{cube}}}{Volume_{\text{sphere}}} = \frac{2}{\pi\sqrt{3}}. \tag{2}$$

Hence, to achieve a desired cubic particle volume fraction V_{cube} in the final model, the temporary spherical packing fraction $V_\circ$ used during seeding must be scaled as

$$V_\circ = \frac{\pi\sqrt{3}}{2} V_{\text{cube}}. \tag{3}$$

The script first places spherical particles at random locations with randomly assigned radii, performing collision checks after each placement and discarding overlaps. As the cumulative particle volume approaches the target, the particle size is adjusted via a prescribed relation to match the LMS-1D particle size distribution. Once the target volume fraction is reached, particle generation stops and three random rotation angles are assigned to each particle.

A second script then checks for intersections with an XY plane through the matrix and writes only the intersecting particles to an OpenSCAD script. OpenSCAD renders cubes with the assigned rotations applied, inscribed within the spheres using the stored parameters.

After defining all particles, a 2D section is taken along the central XY plane of the matrix. This section is saved as a `.dxf` file representing the particles. The script is then rerun with a subtraction operation to generate the complementary matrix geometry, also saved as a `.dxf`.

Under standard stereological assumptions, the areal fraction measured in the 2D section provides an unbiased estimate of the 3D particle volume fraction. This procedure is effective because the target particle volume fraction is first enforced in 3D, and the subsequent mid–plane cut yields a 2D areal fraction that is representative of the true regolith content.

These sketches are imported into Abaqus to create individual parts, assign materials, and assemble the 2D matrix–particle system. The next sections describe how the input parameters required for simulation are extracted, calculated, or estimated.

3. Composite Density Estimation and Regolith Characterization

The LMS-1D datasheet (Exolith Lab, USA) reports two densities: (i) the uncompressed bulk density of loose simulant, which includes inter-particle voids (0.70 g/cm^3), and (ii) a particle density measured by Archimedean displacement in water (2.92 g/cm^3). Neither value is suitable for finite-element inputs in PEEK–regolith composites. The bulk value reflects packing voids rather than the solid phase. The Archimedean value likely overestimates the effective regolith phase density in PEEK because water penetrates open porosity and surface asperities that molten PEEK (high viscosity, different wetting) does not fill during processing. Neither reconciles with the composite densities obtained via the Archimedean displacement (see [6]) method when combined with the known density of PEEK and the pore fraction estimated from micrographs.

This section, therefore, estimates the true particle-phase density and the porosity-free (dense) composite density by combining mass–volume measurements, image-based porosity quantification, and back-calculations across compositions.

3.1. Regolith Simulant Composition

Lunar regolith simulant consists of several oxide phases. Based on the fact sheet, the weight percentages for LMS-1D [14] closely resemble those presented in Table 1.

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Table 1. Weight fractions of different oxides in the material composition of LMS-1D simulant [14].

Oxide	Weight Fraction W_i (%)
SiO_2 (Silica)	46.9
TiO_2 (Rutile)	3.6
Al_2O_3 (Alumina)	12.4
FeO (Wüstite)	8.6
MnO (Manganosite)	0.2
MgO (Periclase)	16.8
CaO (Lime)	7.0
Na_2O (Sodium Oxide)	1.7
K_2O (Potassium Oxide)	0.7
P_2O_5 (Phosphorus Pentoxide)	0.2

3.2. Densities of Oxides (g/cm^3)

Table 2 summarizes room–temperature densities for the relevant oxide constituents used in our calculations. Values are taken from standard handbooks and compilations [14–17], where available.

Table 2. Density of each oxide found in LMS-1D [14–17].

Oxide	ρ_i (g/cm^3)
SiO_2 (Silica)	2.65
TiO_2 (Rutile)	4.23
Al_2O_3 (Alumina)	3.97
FeO (Wüstite)	5.7
MnO (Manganosite)	5.4
MgO (Periclase)	3.58
CaO (Lime)	3.34
Na_2O (Sodium Oxide)	2.27
K_2O (Potassium Oxide)	2.43
P_2O_5 (Phosphorus Pentoxide)	2.39

3.3. Estimation of Regolith Solid Density

To estimate the solid density, we use the equation:

$$\frac{1}{\rho_{\text{regolith}}} = \sum \frac{W_i}{\rho_i} \tag{4}$$

where:

- W_i is the weight fraction of each oxide,
- ρ_i is the density of each oxide.

Assuming no intermediate compositions and ignoring ionic bonding effects, the estimated density of regolith is:

$$\rho_{\text{regolith}} \approx 3.15 \text{ g/cm}^3 \tag{5}$$

3.4. Image-Based Porosity Estimation

A backscattered electron (BSE) SEM image of the 40 wt% PEEK–regolith composite filament (from Azami et al. [6]) is processed to estimate porosity. Using grayscale thresholding and circular masking, the porosity is found to be approximately 1.45%. The measured density of the 40 wt% regolith composite is 1.623 g/cm^3.

Correcting for the porosity using:

$$\rho_{\text{dense}} = \frac{\rho_{\text{measured}}}{1 - \phi_{\text{composite}}} = \frac{1.623}{1 - 0.0145} = 1.6469 \text{ g/cm}^3 \approx 1.647 \text{ g/cm}^3. \tag{6}$$

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3.5. Back-Calculation of Regolith Density

Assuming PEEK has a density of 1.30 g/cm^3 and that the composite contains 40 wt% regolith, the back-calculated (bulk) density of the regolith phase is 2.77 g/cm^3. If the fully dense (pore-free) regolith has a true density of 3.15 g/cm^3 (see above), the intra-particle porosity of the regolith phase ϕ and the corresponding density correction are

$$\phi_{\text{regolith phase}} = \frac{\rho_{\text{true}} - \rho_{\text{bulk}}}{\rho_{\text{true}}} = \frac{3.15 - 2.77}{3.15} \approx 0.121\ (12.1\%). \tag{7}$$

This indicates that the regolith phase contains approximately 12.1% porosity.

3.6. Dense Composite Density Calculation

Using the regolith phase density (2.77 g/cm^3) and PEEK density (1.30 g/cm^3), Table 3 presents the calculated dense composite densities for varying regolith weight fractions.

Table 3. Calculated dense composite densities using regolith phase density (2.77 g/cm^3) and PEEK density (1.30 g/cm^3).

Regolith wt%	Composite Density (g/cm^3)
10	1.373
20	1.454
30	1.546
40	1.650
50	1.770

Note that, to reduce computational complexity in the finite element simulations, regolith particles smaller than 1 µm, comprising approximately 11% of the total regolith volume [14], are excluded from the explicit geometry in Abaqus. Their mechanical contribution is incorporated by homogenizing them with the PEEK matrix. For further information on this homogenization approach and the effective elastic modulus calculations, refer to Section 5.

4. Young's Modulus of LMS-1D: Calculation Using Voigt and Reuss Models

We do not have direct access to the Young's modulus of the regolith particles, either for the bulk material or for individual grains. Moreover, regolith grains are multiphase aggregates with spatially varying properties. For simplicity, each particle is treated as homogeneous with a single, composition-averaged elastic modulus. This assumption is reasonable because the experimental measurements (see [6]) provide bulk moduli at the composite scale, and our goal is to bound effective behavior rather than resolve intragrain heterogeneity. The effective mechanical properties of the regolith particle phase are estimated using classical bounds, primarily the Voigt (iso-strain) [18] upper bound and the Reuss (iso-stress) [19] lower bound, which relate the effective Young's modulus to phase properties and volume fractions.

4.1. Voigt Upper Bound

The Voigt model assumes all phases experience the same strain (parallel arrangement), giving:

$$E_{\text{upper}} = \sum_i V_i E_i, \quad \text{[Voigt; 18]} \tag{8}$$

4.2. Reuss Lower Bound

The Reuss model assumes all phases experience the same stress (series arrangement), yielding:

$$\frac{1}{E_{\text{lower}}} = \sum_i \frac{V_i}{E_i}, \quad \text{[Reuss; 19]} \tag{9}$$

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4.3. Effect of Porosity

Accurate estimation of E_{eff} is essential when comparing defect-free Voigt/Reuss bounds to measured moduli in materials with complex internal porosity. In practical materials, porosity can significantly reduce the effective Young's modulus, as pores act as stress concentrators and limit the material's load-bearing capacity. This mechanical degradation is commonly captured using the following empirical relation:

$$E_{\text{eff}} = E_0(1 - a\phi)^n \quad [20,21], \tag{10}$$

where E_0 is the Young's modulus of the fully dense material, ϕ is the porosity fraction, and a and n are empirical constants reflecting pore morphology and connectivity. For highly irregular and partially connected porosity (e.g., regolith-based composites), literature fits for porous ceramics justify using parameters toward the upper end of typical ranges. In this work we adopt $a = 3.85$ and $n = 4.62$ as representative values [20–23]. Introducing the morphology factor $a > 1$ generalizes the simple $(1 - \phi)^n$ form and improves agreement with data for non-spherical/connected pores [20,22].

4.4. Young's Modulus of Each Oxide in Lunar Regolith Simulant

Based on the literature, the Young's modulus values for the oxide phases are summarized in Table 4 [16,24–29].

Table 4. Young's modulus values for the oxide phases.[1]

Oxide	E_i **(GPa)**
SiO_2 (Silica)	73
TiO_2 (Rutile)	230
Al_2O_3 (Alumina)	380
FeO (Wüstite)	150
MnO (Manganosite)	160
MgO (Periclase)	280
CaO (Lime)	110
Na_2O (Sodium Oxide)	30
K_2O (Potassium Oxide)	20
P_2O_5 (Phosphorus Pentoxide)	10

4.5. Volume Fraction Calculation

The volume fraction V_i is given by:

$$V_i = \frac{W_i/\rho_i}{\sum(W_i/\rho_i)} \tag{11}$$

as per Tables 1 and 2. Computing the denominator:

$$\sum(W_i/\rho_i) = 31.191 \text{ cm}^3/\text{g} \tag{12}$$

Resulting computed V_i for each oxide is summarized in Table 5.

[1] Dense, stable bulk data for Na_2O and K_2O at room temperature are scarce due to their reactivity and hygroscopicity; several studies report single-crystal elastic constants (from which E can be derived), but reliable bulk experimental E values are limited. For FeO (wüstite) and MnO (manganosite), the modulus depends strongly on stoichiometry and defect content, and reported values are frequently inferred from single-crystal elastic constants. For P_2O_5, reported moduli span crystalline polymorphs and phosphate glasses. Where literature variability exists, we provide a representative value.

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Table 5. Volume fractions of different oxides in the material composition.

Oxide	Volume Fraction V_i
SiO_2 (Silica)	0.567
TiO_2 (Rutile)	0.027
Al_2O_3 (Alumina)	0.100
FeO (Wüstite)	0.048
MnO (Manganosite)	0.0012
MgO (Periclase)	0.150
CaO (Lime)	0.067
Na_2O (Sodium Oxide)	0.024
K_2O (Potassium Oxide)	0.0093
P_2O_5 (Phosphorus Pentoxide)	0.0027

4.6. Resulting Young's Modulus Estimates

This section reports the composite stiffness predicted from the micromechanics framework. First, we establish the defect-free bounds using Voigt (iso-strain) and Reuss (iso-stress) estimates and their mean, which together bracket the elastic response of a fully dense material with the measured phase fractions. We then account for processing-induced porosity using an empirical stiffness–porosity relation to obtain the effective Young's modulus at realistic pore levels.

For a porosity-free composite, the classical bounds provide a stiffness interval that reflects the best- and worst-case phase load sharing. The Voigt bound corresponds to perfectly constrained (parallel) phases, while the Reuss bound corresponds to perfectly compliant (series) phases. So, their average serves as a practical central estimate for isotropic polycrystals.

- **Upper Bound (Voigt):** $E_{\text{upper}} = 143.94$ GPa
- **Lower Bound (Reuss):** $E_{\text{lower}} = 89.40$ GPa
- **Average (VRH midpoint):** $E_{\text{avg}} = 116.67$ GPa

To reflect the characteristics of the regolith phase, we reduce the dense modulus by the measured porosity using Equation (10). Here, E_0 is the dense (porosity-free) estimate, ϕ is the porosity fraction obtained from image analysis, and (a, n) capture the sensitivity to pore morphology/connectivity. This converts the idealized stiffness to an effective modulus suitable for comparison with experiments.

With $E_0 = 116.67$ GPa, $a = 3.85$, and $n = 4.62$, we obtain for a representative porosity of $\phi = 0.121$ (12.1%):

$$\begin{aligned} E(\phi{=}0.121) &= 116.67 \times \left(1 - 3.85 \times 0.121\right)^{4.62} \\ &= 116.67 \times (0.534)^{4.62} \\ &\approx \boxed{6.45 \text{ GPa}}. \end{aligned} \tag{13}$$

The calculated value of $E_r = 6.45$ GPa provides a reasonable estimate of the Young's modulus for the regolith particle phase. Again, the true modulus E of a defect-free material lies between the Reuss and Voigt limits. However, in practice, microstructure, phase distribution, and porosity must be considered, since porosity reduces stiffness. This result highlights the severe stiffness knock-down imposed by ∼12% porosity relative to the dense upper-bound behavior, consistent with the strong porosity sensitivity of ceramic materials.

5. Homogenized Matrix Modulus Estimation for Fine Regolith Particles in PEEK

To reduce computational complexity in the finite element simulations, regolith particles smaller than 1 µm, comprising approximately 11% of the total regolith volume [14], are excluded from the explicit geometry in Abaqus. Instead, their mechanical contribution is incorporated by homogenizing them with the PEEK matrix. This section outlines the methodology used to compute the effective elastic modulus of this modified matrix using Voigt and Reuss bounds.

Material properties:

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- Elastic modulus of PEEK (E_m): 1.14 GPa
- Elastic modulus of regolith (E_r): 6.45 GPa
- Fine regolith fraction: 11% of the total regolith volume

The homogenization procedure follows these steps:

1. Converting the target regolith weight fraction (W_{reg}) into a total volume fraction ($V_{\text{reg,orig}}$), using:

$$V_{\text{reg,orig}} = \frac{W_{\text{reg}}/\rho_{\text{reg}}}{(W_{\text{reg}}/\rho_{\text{reg}}) + ((1 - W_{\text{reg}})/\rho_{\text{PEEK}})} \tag{14}$$

where $\rho_{\text{reg}} = 2.77\ \text{g/cm}^3$ and $\rho_{\text{PEEK}} = 1.30\ \text{g/cm}^3$.

2. Separating the regolith volume into coarse (explicitly modeled) and fine (homogenized) components [14]:

$$V_{\text{coarse}} = 0.89 \cdot V_{\text{reg,orig}}, \quad V_{\text{fine}} = 0.11 \cdot V_{\text{reg,orig}} \tag{15}$$

3. Calculating the total matrix volume, which consists of the PEEK matrix and the fine regolith:

$$V_{\text{matrix}} = (1 - V_{\text{reg,orig}}) + V_{\text{fine}} \tag{16}$$

4. Normalizing the fine regolith and PEEK fractions within the matrix:

$$V'_{\text{fine}} = \frac{V_{\text{fine}}}{V_{\text{matrix}}}, \quad V'_{\text{PEEK}} = 1 - V'_{\text{fine}} \tag{17}$$

5. Calculating the homogenized matrix modulus using the Voigt and Reuss bounds:

$$E_V = V'_{\text{PEEK}} \cdot E_m + V'_{\text{fine}} \cdot E_r \tag{18}$$

$$E_R = \left(\frac{V'_{\text{PEEK}}}{E_m} + \frac{V'_{\text{fine}}}{E_r} \right)^{-1} \tag{19}$$

$$E_{\text{avg}} = \frac{E_V + E_R}{2} \tag{20}$$

5.1. Sample Calculation for 10 wt% Regolith

$$V_{\text{reg,orig}} = \frac{0.10/2.77}{(0.10/2.77) + (0.90/1.30)} = 0.0496 \quad (4.96\%) \tag{21}$$

$$V_{\text{fine}} = 0.11 \cdot 0.0496 = 0.00546 \quad (0.55\%) \tag{22}$$

$$V_{\text{matrix}} = 0.9504 + 0.00546 = 0.95586 \quad (95.59\%) \tag{23}$$

$$V'_{\text{fine}} = \frac{0.00546}{0.95586} = 0.00570, \quad V'_{\text{PEEK}} = 1 - 0.00570 = 0.99430 \tag{24}$$

$$E_V = 0.99430 \cdot 1.14 + 0.00570 \cdot 6.45 = 1.1703\ \text{GPa} \tag{25}$$

$$E_R = \left(\frac{0.99430}{1.14} + \frac{0.00570}{6.45} \right)^{-1} = 1.1454\ \text{GPa} \tag{26}$$

$$E_{\text{avg}} = \frac{1.1703 + 1.1454}{2} = 1.1578\ \text{GPa} \tag{27}$$

5.2. Results for 10–50 wt% Regolith

Table 6 reports the homogenized PEEK matrix modulus when blended with fine (smaller than 1 µm) regolith particles, assuming the fines constitute 11% of the regolith volume.

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Table 6. Homogenized matrix modulus considering fine regolith particles (11% of the regolith volume) blended with the PEEK matrix.

Reg. wt%	Volume Fractions (%)				Normalized Matrix Fractions		Homogenized Modulus (GPa)		
	Total Reg.	Coarse	Fine	Matrix	PEEK	Fine Reg.	Voigt	Reuss	Average
10	4.96	4.41	0.55	95.59	0.9943	0.0057	1.1703	1.1454	1.1578
20	10.50	9.35	1.16	90.49	0.9873	0.0127	1.2077	1.1521	1.1799
30	16.75	14.90	1.84	85.10	0.9784	0.0216	1.2549	1.1607	1.2078
40	23.83	21.21	2.62	78.79	0.9667	0.0333	1.3167	1.1721	1.2444
50	31.94	28.43	3.51	71.57	0.9509	0.0491	1.4007	1.1880	1.2943

Incorporating submicron regolith particles into the PEEK matrix via homogenization results in an effective matrix modulus ranging from approximately 1.16 to 1.29 GPa across the studied range of 10–50 wt% regolith. These moduli are used in Abaqus simulations to represent the matrix phase, while only the coarse regolith particles are explicitly modeled. This approach maintains mechanical accuracy while substantially reducing meshing complexity and computational cost.

6. Simulation Results

This section summarizes the finite element simulation results conducted in Abaqus. Each simulation is performed using a CAD model that includes only coarse regolith particles (above 1 µm), while the matrix is assigned an effective modulus computed from the homogenization of PEEK and fine regolith particles (below 1 µm), as detailed in Section 5.

6.1. Mesh Sensitivity Analysis

This section presents the mesh sensitivity analysis conducted to ensure that the finite element results are independent of mesh resolution. The analysis is conducted on the most demanding case, which corresponds to an RVE with the highest regolith loading (50 wt%, approximately 31.94 vol%) and the largest domain size (0.3 mm $\times$ 0.3 mm). This configuration results in the highest particle count and the most extensive matrix–particle interfacial area.

Three mesh resolutions are tested by progressively halving the global seed size: 2 µm, 1 µm, and 0.5 µm. All simulations are run in Abaqus using CAD models containing only coarse regolith particles (above 1 µm). The matrix phase is assigned an effective modulus accounting for the fine particles, as detailed in Section 5.

The table below summarizes the results of the sensitivity analysis. Effective Young's modulus of the composite ($E_{\text{composite}}$) is recorded at each level. Mesh convergence is considered achieved when the change in modulus between successive refinements is below 1%. Based on this criterion, the 1 µm mesh is deemed sufficient and adopted for the rest of the study.

Table 7. Mesh sensitivity analysis for the worst-case RVE with highest regolith content (50 wt%). Simulation parameters and the resulting composite Young's modulus are presented.

CAD File	RVE Dimensions (mm)	Particle Density (%)	Mesh Size (µm)	Element Count	Elongation (%)	E_{regolith} (GPa)	E_{matrix} (GPa)	$E_{\text{composite}}$ (GPa)
15	0.3×0.3	31.94	2	242212	10	6.45	1.29	1.99
15	0.3×0.3	31.94	1	394612	10	6.45	1.29	1.96
15	0.3×0.3	31.94	0.5	990720	10	6.45	1.29	1.95

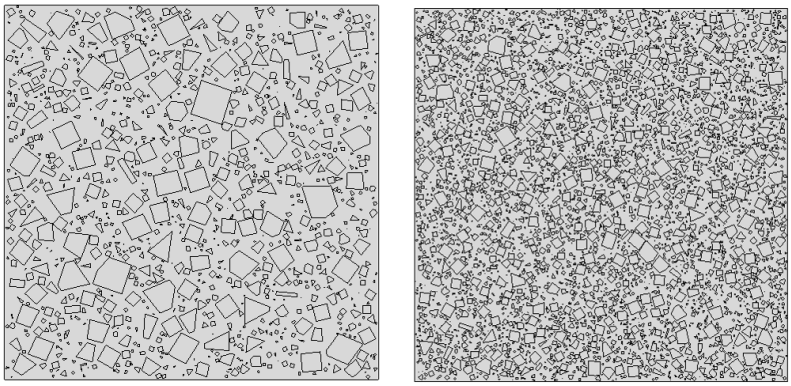
6.2. RVE Size Sensitivity Analysis

This section presents the RVE size sensitivity analysis, which is performed to ensure that the finite element predictions are independent of the RVE domain size. The analysis is conducted using the most demanding configuration, which corresponds to an RVE with the highest regolith content (50 wt%, approximately 31.94 vol%). This setup results in the largest particle count and the most complex matrix–particle interface.

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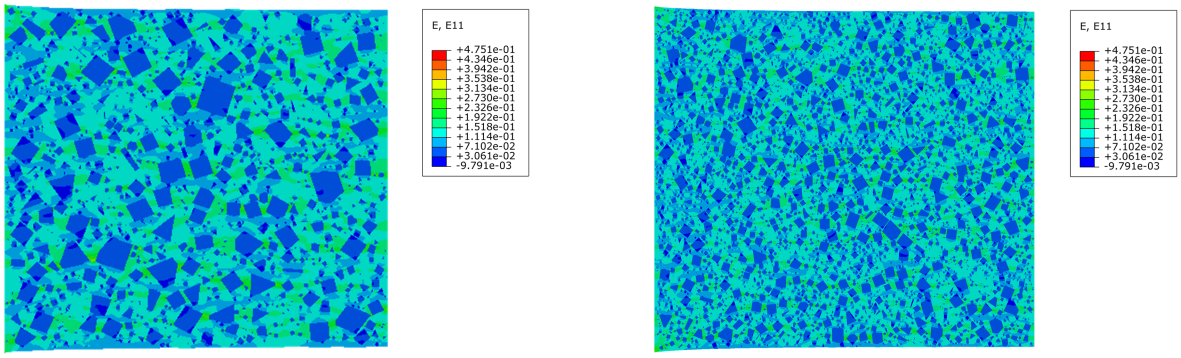
Two RVE sizes are evaluated by doubling the domain dimensions, thereby increasing the area by a factor of four: 0.15 mm (see Figure 1a) and 0.3 mm (see Figure 1b). All simulations are carried out in Abaqus using CAD models that include only coarse regolith particles (larger than 1 µm). The matrix phase is assigned an effective modulus to account for the fine particles, as described in Section 5.



(a) CAD 0 **(b)** CAD 15

Figure 1. CAD models used for the RVE size sensitivity analysis. (a) CAD file No. 0, with an RVE of 0.15 mm × 0.15 mm and a particle loading of 50 wt%. (b) CAD file No. 15, with an RVE of 0.3 mm × 0.3 mm and a particle loading of 50 wt%.

Figure 2a shows the longitudinal strain field E_{11} for the 0.15 × 0.15 mm RVE (CAD file No. 31), and Figure 2b presents the corresponding field for the 0.30 × 0.30 mm RVE (CAD file No. 15). In both cases the matrix carries most of the deformation, with the highest E_{11} concentrated in narrow polymer ligaments and near sharp particle corners, while inclusions remain comparatively strain-shielded. The strain-partitioning pattern is similar between the two RVEs. Enlarging the edge length from 0.15 to 0.30 mm smooths some fine-scale fluctuations yet preserves the topology and areal fraction of high-strain paths. No new banding or emergent length-scale effects appear.



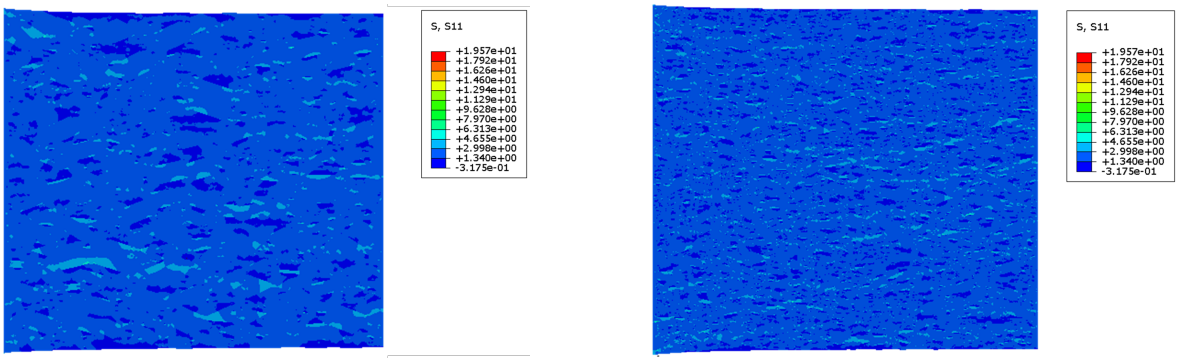
(a) 0.15 mm **(b)** 0.30 mm

Figure 2. Contour plot of the strain component E_{11} obtained from the Abaqus simulation. The results correspond to (a) CAD file No. 0 with an RVE size of 0.15 mm × 0.15 mm and (b) CAD file No. 15 with an RVE size of 0.3 mm × 0.3 mm. In both cases, the mesh size is 1 µm, the matrix and regolith phases have Young's moduli of 1.29 GPa and 6.45 GPa, respectively, and the prescribed elongation is 10 %.

Figure 3a displays the axial stress field S_{11} for the 0.15 × 0.15 mm RVE and Figure 3b for the 0.30 × 0.30 mm RVE under identical macroscopic loading. The stress maps are governed by matrix-borne streaks aligned with the loading direction and by local hot spots at inclusion contacts and re-entrant corners. With the larger RVE these hot spots appear slightly more diffuse due to averaging over a broader particle population, but their intensity levels and spatial frequency remain comparable. The similarity of the stress-bearing skeleton and the limited change in upper-tail stresses indicate convergence of the micro-stress response with respect to RVE size.

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(a) 0.15 mm **(b)** 0.30 mm

Figure 3. Contour plot of the stress component S_{11} obtained from the Abaqus simulation. The results correspond to (a) CAD file No. 0 with an RVE size of 0.15 mm $\times$ 0.15 mm and (b) CAD file No. 15 with an RVE size of 0.3 mm $\times$ 0.3 mm. In both cases, the mesh size is 1 µm, the matrix and regolith phases have Young's moduli of 1.29 GPa and 6.45 GPa, respectively, and the prescribed elongation is 10 %.

Overall, doubling the in-plane RVE dimension from 0.15 to 0.30 mm does not significantly alter strain- and stress-partitioning mechanisms or the homogenized response. Within this size window the microfields are stationary.

In terms of the global elastic modulus, Table 8 shows that the homogenized response is essentially size-insensitive over the tested window: $E_{\text{composite}} = 1.98$ and 1.97 GPa for the 0.15×0.15 and 0.30×0.30 mm RVEs, respectively. The relative change is $\approx 0.5\%$, meeting our convergence criterion of $< 1\%$ between successive sizes. Nevertheless, we adopt the larger 0.30 mm RVE for all subsequent analyses as a conservative choice. The larger domain reduces potential finite-size bias, better averages microstructural clustering, and yields smoother tail statistics in the strain/stress fields. In short, the global modulus has converged, but the 0.30 mm RVE offers greater statistical representativeness at acceptable computational cost and is therefore selected going forward.

Table 8. RVE size sensitivity analysis for the highest regolith content (50 wt%). Simulation parameters and the resulting composite Young's modulus are presented.

CAD File	RVE Dimensions (mm)	Particle Density (%)	Mesh Size (µm)	Element Count	Elongation (%)	E_{regolith} (GPa)	E_{matrix} (GPa)	$E_{\text{composite}}$ (GPa)
0	0.15×0.15	31.94	1	56,753	10	6.45	1.29	1.98
15	0.30×0.30	31.94	1	394,612	10	6.45	1.29	1.97

6.3. Effect of Regolith Loading

This section examines the influence of regolith loading on the composite's mechanical behavior. The analysis is carried out using an RVE of dimensions 0.3×0.3 mm with a mesh size of 1 µm. In alignment with the experimental setup (see [6]), five regolith loadings are considered: 10 wt% (4.96 vol%), 20 wt% (10.50 vol%), 30 wt% (16.75 vol%), 40 wt% (23.83 vol%), and 50 wt% (31.94 vol%). Representative CAD geometries for these cases (CAD files 1, 6, 7, 10, and 15) are shown in Figure 4.

Representative strain (E_{11}) and stress (S_{11}) maps for 10, 20, 30, 40, and 50 wt% regolith are shown in Figures 5b–6b, 5c–6c, 5d–6d, 5e–6e, and 5f–6f, respectively. Together they reveal a consistent, matrix-dominated load-transfer mechanism with progressively tighter polymer ligaments as the inclusion content increases.

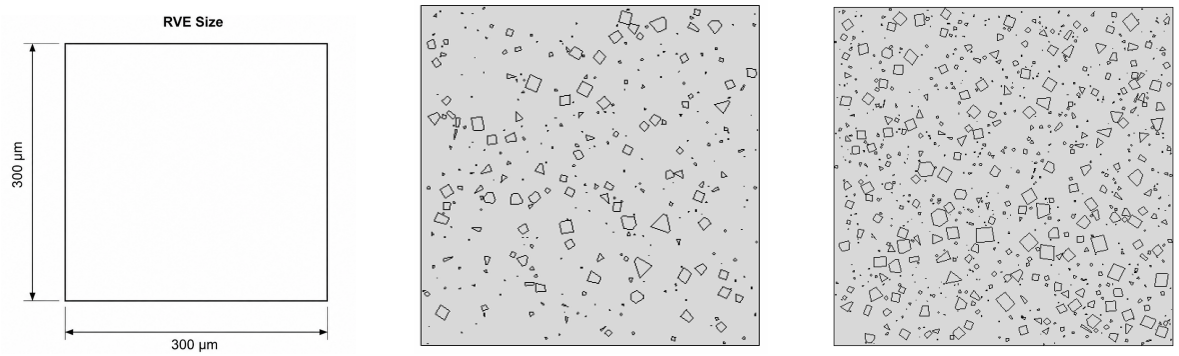
10 wt% (CAD 1). In Figures 5b and 6b, inclusions are well separated within a continuous polymer matrix. The E_{11} field is smooth at the mesoscale, with small, localized low-strain halos around particles and mild strain amplification in intervening ligaments. The S_{11} map shows sparse, short stress streaks aligned with loading and limited hot spots at re-entrant corners.

20 wt% (CAD 6). Figures 5c and 6c show reduced inclusion spacing due to higher particle loading. Strain-shielded zones expand and begin to overlap, producing more tortuous high-strain paths through

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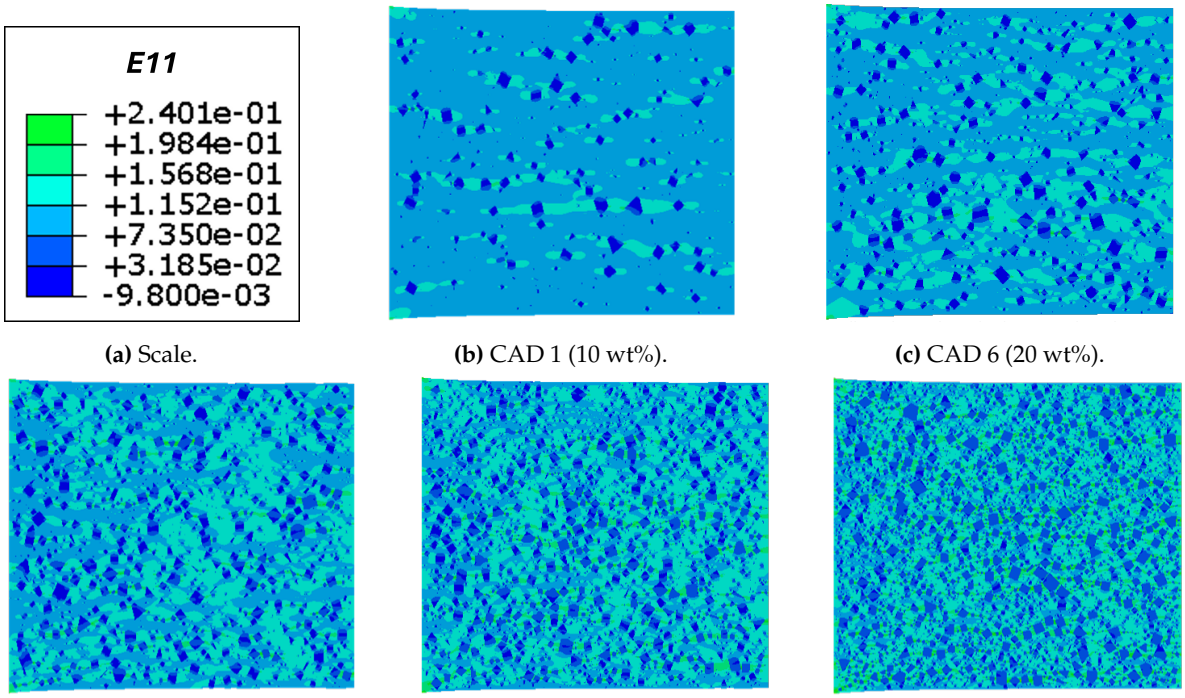
the matrix. The S_{11} field exhibits more numerous hot spots and longer streaks, especially where two particles approach within a ligament thickness.



(a) RVE size (300 μm $\times$ 300 μm) **(b)** CAD 1 (10% wt) **(c)** CAD 6 (20% wt)

(d) CAD 7 (30% wt) **(e)** CAD 10 (40% wt) **(f)** CAD 15 (50% wt)

Figure 4. Representative CAD geometries generated for finite-element analyses, corresponding to regolith loadings of (a) RVE size, (b) 10 wt%, (c) 20 wt%, (d) 30 wt%, (e) 40 wt%, and (f) 50 wt%.



(a) Scale. **(b)** CAD 1 (10 wt%). **(c)** CAD 6 (20 wt%).

(d) CAD 7 (30 wt%). **(e)** CAD 10 (40 wt%). **(f)** CAD 15 (50 wt%).

Figure 5. Representative E_{11} strain microfields for regolith loadings of 10, 20, 30, 40, and 50 wt%, shown together with the corresponding color scale. Increasing particle content produces progressively narrower polymer ligaments and stronger strain partitioning.

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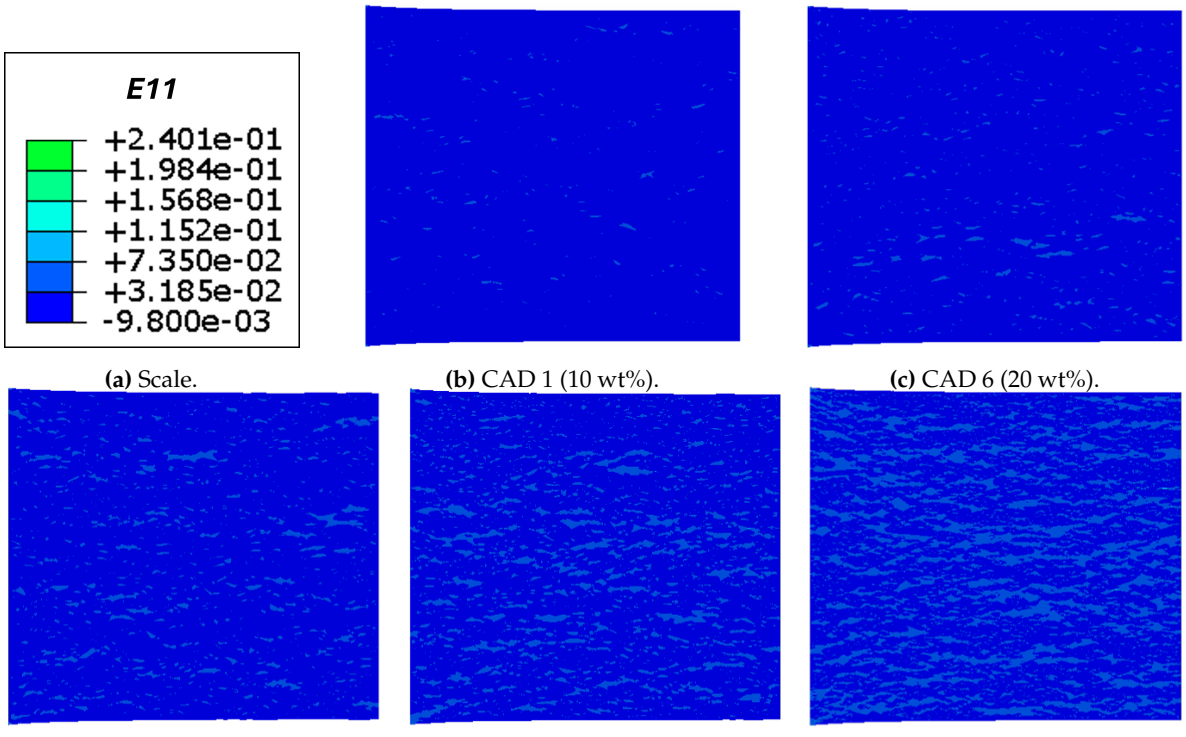
30 wt% (CAD 7). In Figures 5d and 6d, quasi-continuous clusters appear. Polymer ligaments are thinner and more anisotropic, so E_{11} localization becomes more segmented. Stress hot spots concentrate at multi-particle junctions and along narrow ligaments. The stress-bearing skeleton is denser, and the spacing between hot regions is reduced relative to 20 wt%.

40 wt% (CAD 10). Figures 5e and 6e show common particle clusters and a thinned polymer network. Connectivity is reduced, but the matrix remains continuous. The E_{11} maps display extensive low-strain zones with high-strain channels threading the remaining ligaments. In S_{11}, elongated streaks and near-contact hot spots dominate, with higher intensity and frequency than at 30 wt%.

50 wt% (CAD 15). As seen in Figures 5f and 6f, ligaments are very narrow and strain is strongly partitioned. Inclusions remain low-strain while the matrix accommodates most deformation along a fine network of high-E_{11} channels. The S_{11} field is saturated with short-wavelength streaks and hot spots reflecting frequent near-contacts and sharp corners. Despite this micro-level intensification, the macroscale stiffness gain is moderate because the regolith–matrix modulus contrast is only of order unity. The composite remains matrix-controlled even at this highest loading.

Trend Synthesis

Across 10 $\rightarrow$ 50 wt% (Figures 5 and 6), (i) the matrix carries the majority of strain while inclusions are strain-shielded; (ii) the density and intensity of S_{11} hot spots increase with regolith loading due to thinner ligaments and more frequent near-contacts; and (iii) the stress-bearing skeleton becomes finer and more continuous, with no obvious shear-band organization. This explains the change in homogenized stiffness alongside progressively heavier local stress tails.



(a) Scale. **(b)** CAD 1 (10 wt%). **(c)** CAD 6 (20 wt%).

(d) CAD 7 (30 wt%). **(e)** CAD 10 (40 wt%). **(f)** CAD 15 (50 wt%).

Figure 6. Representative S_{11} stress microfields for regolith loadings of 10, 20, 30, 40, and 50 wt%, shown together with the color scale. Stress hot spots intensify and become more frequent with higher filler contents due to thinner ligaments and increased particle proximity.

Table 9 includes three independent RVEs per composition to demonstrate reproducibility. The small scatter ($\leq$ 0.02 GPa) indicates that the 0.30 $\times$ 0.30 mm RVE captures the statistics of the microstructure with good repeatability. Under the idealized assumptions (linear-elastic phases, perfect particle–matrix bonding, and no process defects), the simulations exhibit a smooth, monotonic increase

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in the homogenized modulus with rising regolith content. Table 10 then compares these predictions against measurements from additively manufactured coupons produced in [6].

Table 9. Finite-element predictions of composite Young's modulus for RVEs with different regolith particle densities. Three statistically independent RVEs are included per composition to demonstrate reproducibility.

CAD File	RVE Dimensions (mm)	Particle Density (vol%)	Particle Density (wt%)	Mesh Size (µm)	Element Count	Elongation (%)	E_{regolith} (GPa)	E_{matrix} (GPa)	$E_{\text{composite}}$ (GPa)
1	0.3×0.3	4.96	10	1	335,268	10	6.45	1.16	1.27
2	0.3×0.3	4.96	10	1	315,282	10	6.45	1.16	1.27
3	0.3×0.3	4.96	10	1	328,974	10	6.45	1.16	1.28
4	0.3×0.3	10.50	20	1	360,310	10	6.45	1.18	1.38
5	0.3×0.3	10.50	20	1	336,102	10	6.45	1.18	1.39
6	0.3×0.3	10.50	20	1	359,858	10	6.45	1.18	1.39
7	0.3×0.3	16.75	30	1	346,902	10	6.45	1.21	1.54
8	0.3×0.3	16.75	30	1	355,626	10	6.45	1.21	1.53
9	0.3×0.3	16.75	30	1	348,202	10	6.45	1.21	1.52
10	0.3×0.3	23.83	40	1	360,466	10	6.45	1.24	1.71
11	0.3×0.3	23.83	40	1	358,802	10	6.45	1.24	1.70
12	0.3×0.3	23.83	40	1	351,312	10	6.45	1.24	1.71
13	0.3×0.3	31.94	50	1	396,622	10	6.45	1.29	1.96
15	0.3×0.3	31.94	50	1	394,612	10	6.45	1.29	1.97
16	0.3×0.3	31.94	50	1	387,558	10	6.45	1.29	1.99

Table 10. Mean FE-predicted modulus with sample standard deviation (three RVEs per composition), experimental modulus from AM coupons, and percent difference computed as $(\overline{E}_{\text{FE}} - E_{\text{Experiment}})/E_{\text{Experiment}} \times 100$.

wt%	vol%	$\overline{E}_{\text{FE}}$ (GPa)	$E_{\text{Experiment}}$ (GPa)	Diff. (%)
10	4.96	1.27 ± 0.01	1.23	3.50
20	10.50	1.39 ± 0.01	1.31	5.88
30	16.75	1.53 ± 0.01	1.40	9.29
40	23.83	1.71 ± 0.01	1.55	10.13
50	31.94	1.97 ± 0.02	1.27	55.35

The AM measurements show acceptable agreement with the simulations up to 40 wt%. As the particle volume fraction increases from 5 to 24 vol% (10–40 wt%), the maps indicate that (i) inclusions remain largely strain-shielded, (ii) the matrix develops a progressively finer network of high-E_{11} channels, and (iii) S_{11} hot spots become more numerous but remain discontinuous. The E_{11}/S_{11} maps (Figures 5b–6f and geometries in Figure 4) explain this trend. This morphology preserves a continuous, load-bearing polymer skeleton, so the global modulus increases modestly and the idealized FE agrees reasonably with experiment. Over this range the stiffening is gradual, a trend which is captured both in simulation and experiment. Percent differences between simulation and experiment are approximately 3.5%, 5.9%, 9.3%, and 10% at 10–40 wt%, respectively; these small but growing residual gaps are consistent with the observed increase in defect content and the progressive loss of matrix continuity as regolith loading rises, as corroborated by density measurements and microstructural analyses in [6], and with the simplifying assumptions in the simulations.

The results deviate strongly at 50 wt% (experiment 1.27 GPa vs. FE 1.97 GPa, a $\sim$ 55% gap). The simulated value shows continued gradual increase, while in the experimental dataset there is a drop in E between 40 and 50 wt% [6]. To attempt to capture this reversal in trend, defect-aware and single-printed-layer modeling is adopted, presented in the next subsections.

6.4. Defect-Aware Tiling Strategy

A 0.9 × 0.9 mm region of the SEM image is shown in Figure 7, also indicating a partitioning into a 3 × 3 grid of 0.3 × 0.3 mm tiles. As can be seen, different 0.3 × 0.3 mm patches in the same micrograph exhibit very different defect populations, so directly reproducing the exact defect field from SEM in a single 0.3 × 0.3 mm RVE is not reliable.

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Two paths are considered: (i) a much larger single RVE that statistically averages the heterogeneity, which is computationally prohibitive at these particle counts; or (ii) a defect-aware tiling strategy that preserves spatial variability, described here.

A corresponding RVE is assigned to each tile in which the observed local defects are embedded. Each tile is analyzed in the small-strain elastic regime to obtain its axial modulus E_{ij}. The effective stiffness of the entire 0.9×0.9 mm region is estimated using the classical Voigt and Reuss bounds [18,19] and their arithmetic mean (Hill average).

$$E_{\text{Voigt}} = \frac{1}{9} \sum_{i=1}^{3} \sum_{j=1}^{3} E_{ij}, \tag{28}$$

$$E_{\text{Reuss}} = \left(\frac{1}{9} \sum_{i=1}^{3} \sum_{j=1}^{3} \frac{1}{E_{ij}} \right)^{-1}, \tag{29}$$

$$E_{\text{Hill}} = \tfrac{1}{2} \left(E_{\text{Voigt}} + E_{\text{Reuss}} \right). \tag{30}$$

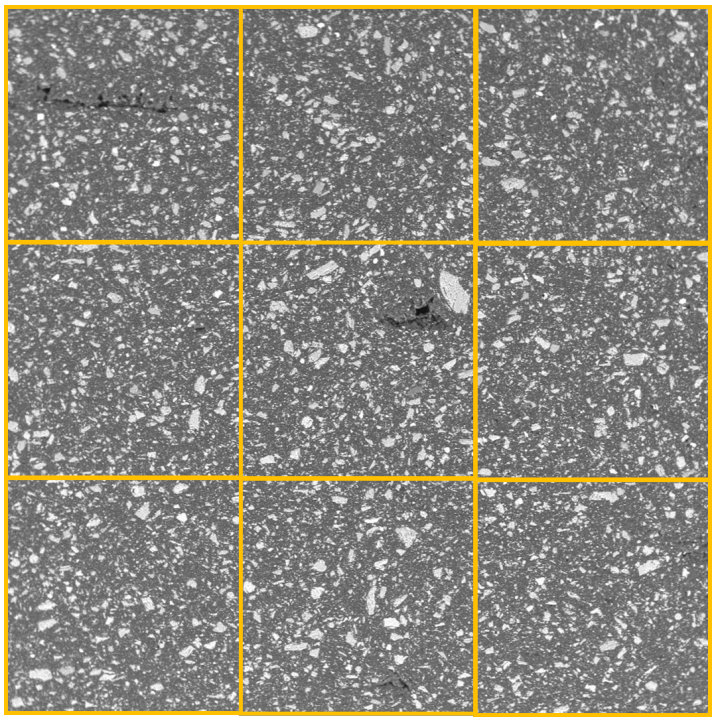


Figure 7. A 0.9×0.9 mm region from the SEM micrograph of the PEEK/LRS50 specimen, subdivided into a 3×3 array of 0.3×0.3 mm tiles. Each tile is used to construct a defect-aware RVE that captures local defect variability [6].

6.4.1. Application to a Representative 3×3 Set

The proposed defect-aware tiling methodology is applied to the region shown in Figure 8. The selected area is first partitioned into nine 0.3×0.3 mm tiles. Defects identified within each tile are then transferred to the corresponding RVE geometry, and the effective elastic modulus of each tile is determined through finite element analysis. The resulting tile moduli (GPa) are taken as

$$\left[1.90,\ 1.97,\ 1.97;\ 1.95,\ 1.60,\ 1.96;\ 1.97,\ 1.93,\ 1.95 \right],$$

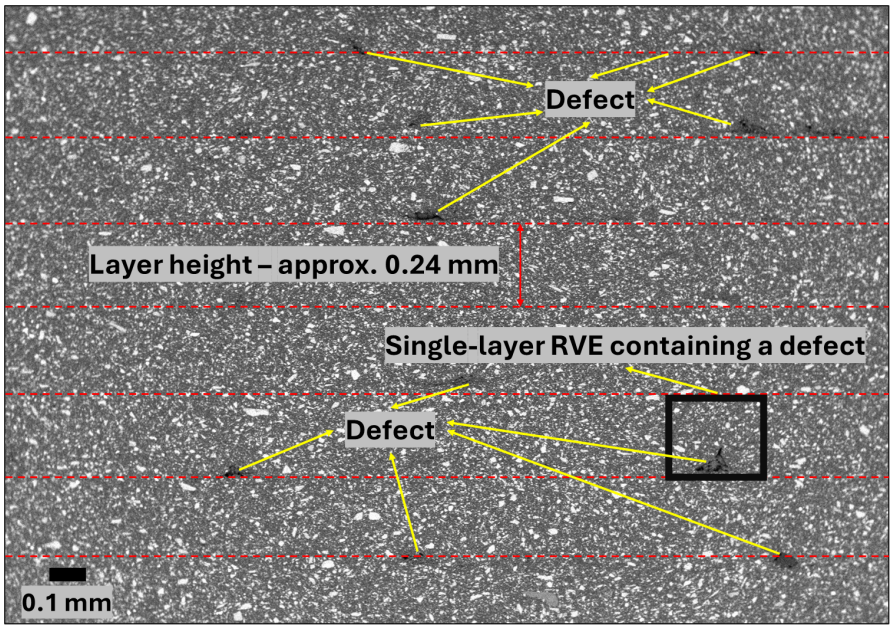
from which the following region-level estimates are obtained:

$$E_{\text{Voigt}} = 1.911 \text{ GPa}, \quad E_{\text{Reuss}} = 1.903 \text{ GPa}, \quad E_{\text{Hill}} = 1.907 \text{ GPa}.$$

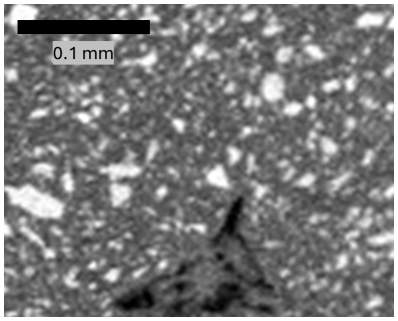
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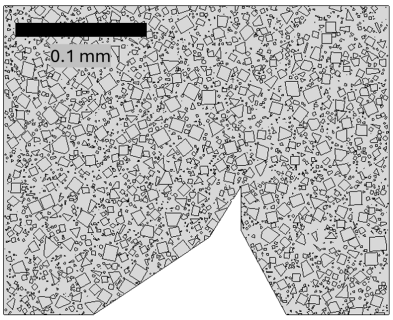
The results reveal a clear correlation between local defect severity and the predicted elastic modulus. Tile 5 exhibits the lowest modulus (1.60 GPa), corresponding to the large defect visible in the central tile of Figure 8. The second-lowest modulus is observed for Tile 1 (1.90 GPa), which also contains a noticeable defect. In contrast, tiles with minimal visible defects exhibit moduli close to 1.95–1.97 GPa. These findings indicate that the defect-aware tiling approach is capable of capturing the local stiffness reduction associated with defect-rich regions. However, these values remain significantly above the experimental modulus of 1.27 GPa at 50 wt%. It is therefore concluded that, although realistic heterogeneity and weak tiles are introduced by the tiling approach, the full reduction measured in the printed coupons is not captured by this method alone.



(a) Larger SEM field of the 50 wt% sample. Dashed red lines indicate printed layers (layer height ≈ 0.24 mm); the black box marks the single-layer RVE location.



(b) Zoomed tile with a representative defect.



(c) Single-layer geometry adopted from the SEM.

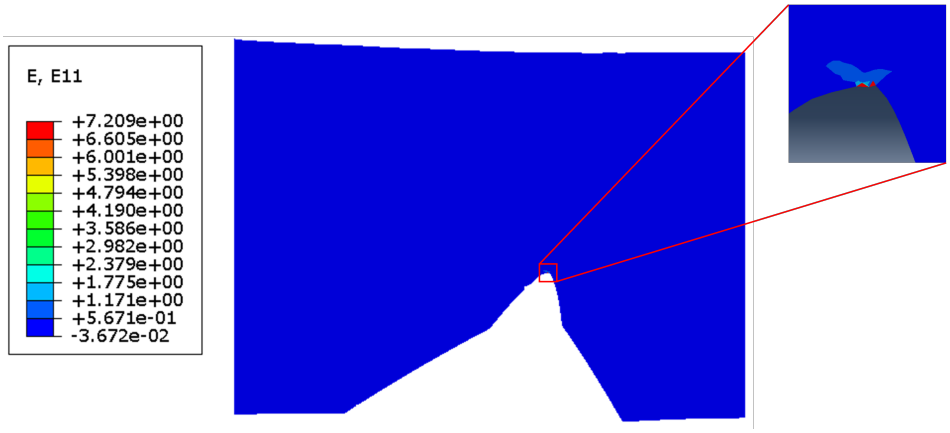
Figure 8. Defect-aware single-layer RVE construction. (a) Larger SEM field of the PEEK/LRS50 sample with printed-layer boundaries marked by dashed red lines; the black box denotes the selected tile. (b) Zoomed-in view of the representative defect. (c) Idealized single-layer geometry extracted from the SEM image for finite-element modeling [6].

To further interrogate the discrepancy, a complementary approach is adopted: the worst tile is selected (i.e., the tile containing the defect with the largest perpendicular extent relative to the loading direction). Figure 8a shows a larger SEM field of the PEEK/LRS50 sample with the printed-layer boundaries and the selected tile (black box) used for the single-layer RVE. Notably, the spacing between the locations of large defects along the y-direction closely matches the printed layer height, suggesting a strong correlation with the layer-wise deposition process and potentially indicating weakened inter-layer bonding. A single-printed-layer RVE with in-plane dimensions 0.3 mm (along x, parallel to the

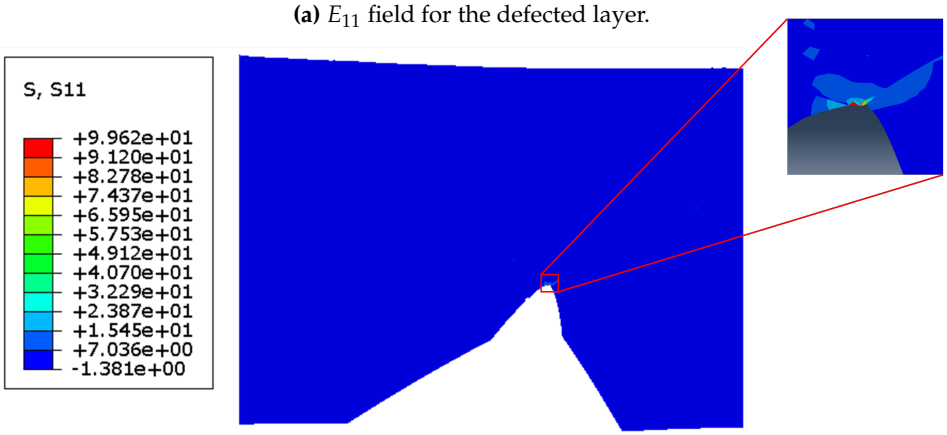
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load) and 0.24 mm (along y, equal to one printed-layer thickness) is constructed. Figure 8 shows the zoomed-in SEM defect and the corresponding idealized geometry used in the model, and Figure 9 presents the resulting E_{11} and S_{11} fields. For this tile, the homogenized modulus decreases from $\sim$1.96 GPa (defect-free) to $\sim$1.45 GPa, confirming the pronounced stiffness penalty when the defect height-to-tile height ratio, $h_{\text{defect}}/h_{\text{tile}}$, is large.



(a) E_{11} field for the defected layer.



(b) S_{11} field for the defected layer.

Figure 9. Computed microfields for the defected single-layer RVE. (a) Axial strain field E_{11} showing strong localization within the narrow ligament surrounding the defect. (b) Axial stress field S_{11} illustrating pronounced stress concentrations and reduced effective stiffness at the layer scale.

6.4.2. Single-Printed-Layer Response

With a large, crack-like void inside one printed layer (Figures 8 and 9), strain localizes within the thin polymer ligament around the void. Load is then transmitted through a few narrow ligaments, and the layer effectively behaves like two stiff pieces joined by a soft seam. The narrow ligament exhibits strong stress concentration and a locally reduced effective modulus, which lowers the overall (RVE-level) stiffness. In the representative case of Figures 9a–9b, the RVE modulus falls from about 1.96 GPa (no defect) to about 1.45 GPa, showing how a single tall void, representing a significant proportion of the layer thickness ($\approx$ 41% in this example), can strongly reduce stiffness.

6.4.3. Part-Level Implications

In the printed part, many such interfacial voids are distributed across successive layers. Although the total porosity at 50 wt% is modest, the defect morphology is likely highly detrimental. Elongated voids and partially starved welds occurring along bead–bead and layer–layer boundaries (as shown in Figure 8a), together with local crystallinity variations and early particle–matrix micro-debonding, may create a network of compliant interphases. As these weak regions become connected across multiple layers, the effective load path must repeatedly traverse them, plausibly leading to a network-controlled stiffness that is substantially lower than that predicted by defect-free RVEs or tile-wise

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averaging. This network effect offers a plausible explanation for why the defect-aware tiling approach still overpredicts E for the whole coupon, whereas the single-printed-layer analysis captures how tall, crack-like interfacial defects can drive the modulus toward the experimental value at 50 wt%.

A broader mechanistic interpretation of this drop is provided in the following section.

7. Mechanistic Origin of the Modulus Drop at 50 wt% Under In-Plane Loading

7.1. Observation and Problem Statement

For the 50 wt% (volume fraction $\phi \approx 0.32$) PEEK/regolith composite loaded parallel to the printed layers, a defect-agnostic finite-element (FE) model predicts $E_{\mathrm{FE}} = 1.97 \pm 0.02$ GPa, whereas tensile tests yield $E_{\mathrm{Experiment}} = 1.27$ GPa. Micrographs and porosimetry indicate a modest total porosity ($\phi \approx 0.075$), but the defects seem to be strongly interfacial and crack-like (elongated voids aligned with bead–bead and layer–layer boundaries and polymer-starved welds). It should be noted that the porosity value of $\phi \approx 0.075$ refers to the bulk porosity of the composite, measured experimentally in [6]. By contrast, the 12.1% (please refer to 3.5) value refers specifically to the porosity of the regolith phase within the composite. This value was estimated by (i) measuring part porosity for the 40 wt% sample via image analysis (1.45%), (ii) combining that measurement with the Archimedean composite density to infer the fully dense composite density, (iii) using PEEK's density (1.30 g/cm^3) to back-calculate the in-composite regolith-phase density, and (iv) comparing that density to the fully dense regolith density computed from its oxide constituents and weight fractions. The resulting ratio yields a regolith-phase porosity of 12.1%.

Adding a few large defects to the RVEs, through the "tiling method" explained earlie,r lowers the prediction only to about 1.90 GPa. However, for a worst-case tile containing a single 0.24 mm printed layer with a major defect, it is reduced to about 1.45 GPa.

The measured 1.27 GPa therefore requires additional interfacial compliance beyond isolated pores or a single extreme defect. The tiling method underpredicts the stiffness loss because it treats the heterogeneous microstructure as a set of locally averaged tiles, so weak regions remain diluted by stiffer neighboring tiles in the Voigt, Reuss, or Hill homogenization rather than acting as a continuous damage path. At 50 wt% regolith, the modulus drop is instead governed by connected defect network, polymer-starved welds, and early particle–matrix debonding that create a network-controlled, series-compliance response across layers, which tile-wise averaging cannot capture.

7.2. How Interfaces Could Dominate Even When Load Is Parallel to Layers

The main mechanisms that could explain the significant drop in Young's modulus are summarized below. The aim is to indicate where the basic FE assumptions become limiting at 50 wt%.

1. **Parallel loading does not ensure uniform strain.**
   If interlayer welds were fully consolidated and defect-free, strain sharing would be expected to remain nearly uniform. At 50 wt%, however, the welds are likely polymer-starved and may exhibit partial debonding, so adjacent roads may transfer load inefficiently. Numerous defects appear at interfaces between successive layers, suggesting weakened interlayer bonding, reduced adhesion, and non-uniform strain transfer. Collectively, these features imply that interlayer regions may act as early-compliant seams that disrupt load continuity even at small strains.
2. **Porosity fraction is modest; porosity shape is harmful.** Although the overall porosity is modest ($\phi \approx 7.5\%$), the elongated interfacial voids likely behave more like short, crack-like defects than conventional rounded pores. Their geometry can amplify local strain at weld roots, interrupt stress transfer between adjacent roads, and may even trigger interfacial slip or opening at very small global strains. When such voids are aligned along bead or layer boundaries, the resulting crack-like porosity can impose a disproportionately large stiffness penalty, consistent with the localized strain and stress patterns observed in Figure 9.

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3. **Early particle–matrix micro-debonding lowers the initial tangent modulus** [2] At high filler loading, angular particles and residual stresses likely hinder wetting. Nanoscale gaps at particle–matrix interfaces may open at very small strains, creating a compliant annulus around particles while both phases remain nominally elastic. Since welds already concentrate strain, micro-debonding is expected to initiate and grow preferentially along these regions, which lowers the effective stiffness relative to models that assume perfect bonding. This mechanism specifically reduces the initial tangent modulus because the compliant interfaces begin to deform earlier than the bulk matrix, even under small applied loads, causing the stress–strain curve to have a lower slope at the onset of loading.
4. **Local losses of crystallinity and orientation at welds.** Higher melt viscosity at 50 wt% may reduce the time available for chain interdiffusion and sintering at bead contacts, as noted by Chuang et al. [30], who observed that increased solid loading can raise melt viscosity. Faster heat extraction near interfaces may further suppress crystallinity. Because PEEK stiffness depends on both, weld zones are likely softer than road interiors. The gauge then behaves like stiff road segments separated by softer seams, which adds series compliance through interfacial slip.
5. **Connectivity of weak regions.** At $V_{\text{regolith}} \approx 0.32$, the compliant interphases, starved welds, imperfect particle shells, and strings of crack-like voids may approach a percolation threshold. Once a weak network forms, every load path is likely to cross it, so the macroscopic response becomes governed by the soft network rather than by the volume-average properties of the phases. The effective modulus becomes series-controlled (i.e., weak segments lie in series along the load path, so their compliances add and the softest/thinnest ligaments dominate). Even thin ligaments with locally low stiffness can dominate the overall response when they connect from one side of the gauge to the other. Practically, this may manifest as (i) a rise in the area fraction and contiguity of high-S_{11} zones at weld roots, (ii) reduced minimum ligament width along layer boundaries, and (iii) the emergence of a single connected weak cluster across the field of view. This connectivity provides a plausible explanation for the non-monotonic trend in stiffness: it rises with filler up to about 40 wt%, then drops at 50 wt% despite the higher solid fraction predicted by perfectly bonded micromechanics.

7.3. Synthesis

Up to 40 wt% regolith, the experimental modulus remains close to defect-free FE predictions. Including modest, isolated defects would likely improve agreement and is expected to have only a limited influence in this range. At 50 wt% a substantial gap appears: the FE response decreases only slightly, from about 1.97 GPa to about 1.90 GPa when defects are added in an averaged sense, whereas the experiment gives 1.27 GPa. The frequency of tall interfacial defects appears high, with several large defects present in each layer. Interlayer welds are likely weak, so load may not be transferred uniformly between layers. These effects lower the modulus of individual layers and hinder uniform load transfer through the thickness. Particle–matrix boundaries are likely polymer-starved at 50 wt%, and interfacial opening may initiate at relatively low strains.

Overall, the data and observations suggest a transition to an interface-controlled regime in which starved welds, early interfacial debonding, and local losses of crystallinity and orientation combine to form a connected, slip-compliant network along layer boundaries. The simplifying assumptions in the model, especially the neglect of the nonlinear and connected effects of defect morphology, imperfect bonding, and spatial heterogeneity, likely contribute to overprediction when the tensile axis is parallel to layers.

8. Conclusione and Future Work

A hierarchical modeling framework, comprising (i) a homogenized matrix with fine-particle estimates, (ii) RVE-based defect-free composite models, and (iii) defect-aware RVEs was employed

[2] The instantaneous slope $d\sigma/d\varepsilon$ at $\varepsilon \to 0$, which is more sensitive to early compliance mechanisms than Young's modulus.

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to analyze composite behavior across particle–matrix ratios. The defect-free models captured the measured stiffness trends with acceptable accuracy up to $\sim$ 40 wt% LRS. The model–experiment divergence increased from $\sim$ 3.50% at 10 wt% to $\sim$ 10.13% at 40 wt%, consistent with acceptable but rising defect density, and then jumped to $\sim$ 55.35% at 50 wt%. This large mismatch at 50 wt% is most plausibly attributed to the combined effects of a higher critical void fraction and weak/partly debonded inter-bead interfaces. The simulations quantify how tall interfacial defects and polymer-starved welds may suppress load transfer, even under in-plane loading, producing a clear modulus knock-down attributable to realistic AM defect ensembles. Once a weak, percolating network forms, all load paths traverse it, so the macroscopic response is likely governed by that soft network rather than by volume-average phase properties. The effective modulus might become series-controlled (compliant segments lie in series along the load path, their compliances may add, and the thinnest or softest ligaments could potentially dominate). Practically, this appears as (i) an increased area fraction and contiguity of high-stress zones at weld roots, (ii) reduced minimum ligament width along layer boundaries, and (iii) the emergence of a single connected weak cluster across the field of view. This connectivity provides a plausible explanation for the non-monotonic trend: stiffness rises with filler to $\sim$ 40 wt%, then falls at 50 wt%.

As future work, it will be valuable to extend the present 2D midsection analysis to statistically representative 3D RVEs that capture through-thickness weld topology, bead curvature, and defect connectivity. Fully 3D simulations would help avoid potential scaling discrepancies associated with inherent differences between 2D and 3D representations. In addition, future studies should investigate more physically representative boundary conditions beyond the current fixed constraint at the left edge of the specimen. Implementing periodic boundary conditions in both the in-plane and through-thickness directions, or using mixed periodic-kinematic constraints, would better emulate an infinite repeating microstructure and reduce artificial stress concentrations introduced by fixed boundaries. Such boundary conditions are more appropriate for printed rasters embedded in a continuous material field.

The transition to 3D RVEs will likely require advanced numerical strategies and/or higher-performance computing resources, such as GPU-based solvers, reduced-order modeling, or adaptive mesh refinement, to manage the increased element count and defect complexity. Incorporating experimentally derived pore size and shape distributions from micro-CT, together with interface traction-separation laws calibrated from fracture testing or micro digital image correlation (micro-DIC), would enable more realistic predictions. In parallel, advanced characterization tools such as FE-SEM, FIB-SEM tomography, ion-polished cross-sections, and nanoindentation modulus mapping could be used to assess particle-matrix interfacial quality and interlayer bonding. Complementary Z-direction tensile testing may further quantify interface strength.

Beyond this, it will be useful to develop process-to-structure models for filament extrusion and cooling, bead-bead sintering and neck growth, and transient chamber thermal fields. Calibration using infrared thermography and embedded thermocouples during FFF could then be used to predict the resulting stiffness and strength. Moreover, it will be important to propagate variability in particle size, morphology, porosity, and interface toughness into the mechanical response, thereby establishing conservative design allowables for lunar service. Finally, compression-molded parts are expected to show closer agreement with the present simulation results, since printing-induced interfacial weaknesses and crack-like defects should be strongly reduced. Future work can therefore focus on compression molding of the parts, followed by direct comparison of the experimental response with the predicted mechanical behavior.

Conflicts of Interest: On behalf of all authors, the corresponding author states that there is no conflict of interest.

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